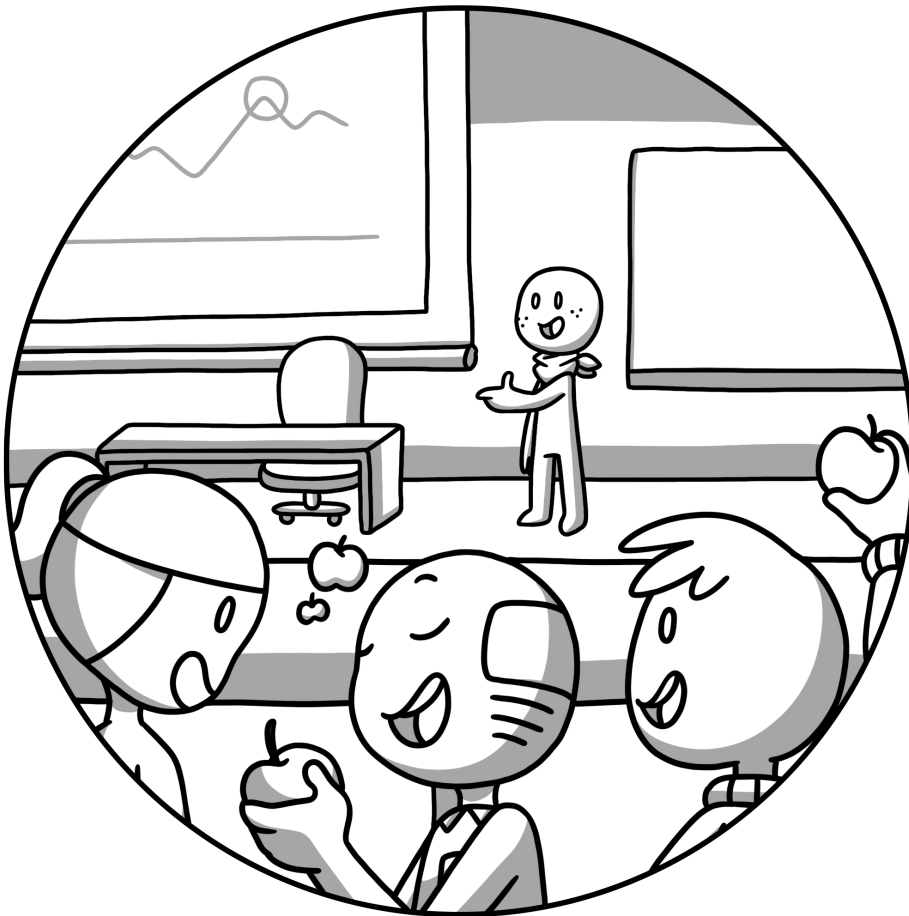


Central Tendency



COMIXPLAIN

This comic was created as part of the Comixplain research project, funded by the University of Applied Sciences St. Pölten (USTP) through the Innovation Call 2022.

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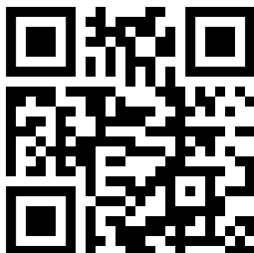
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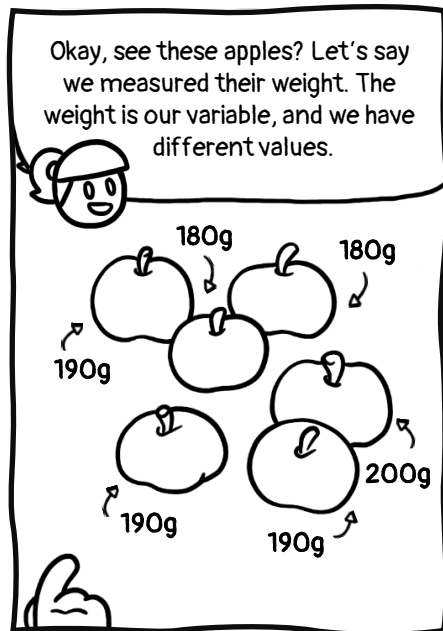
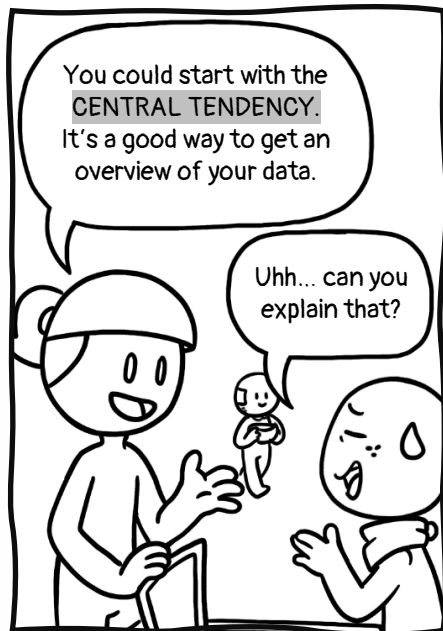
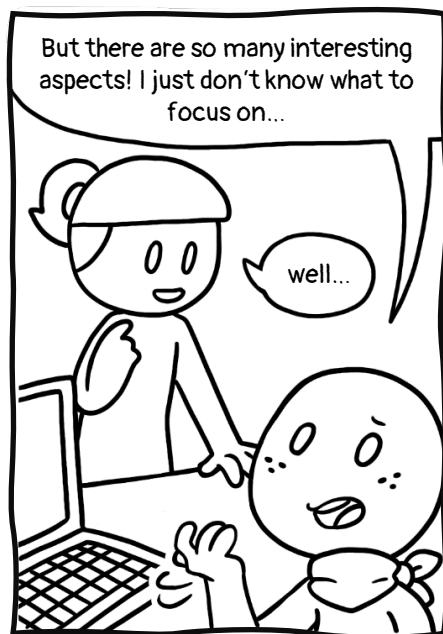
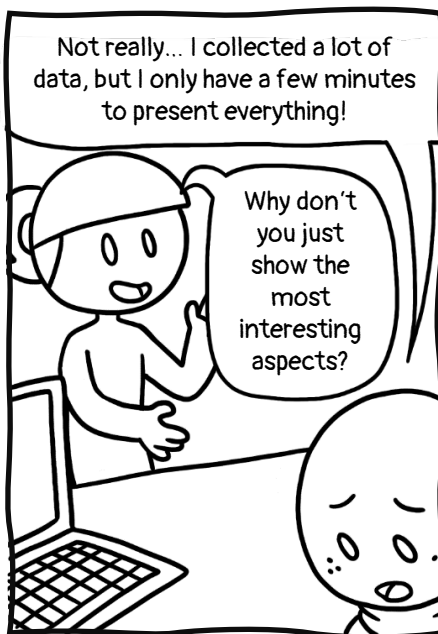
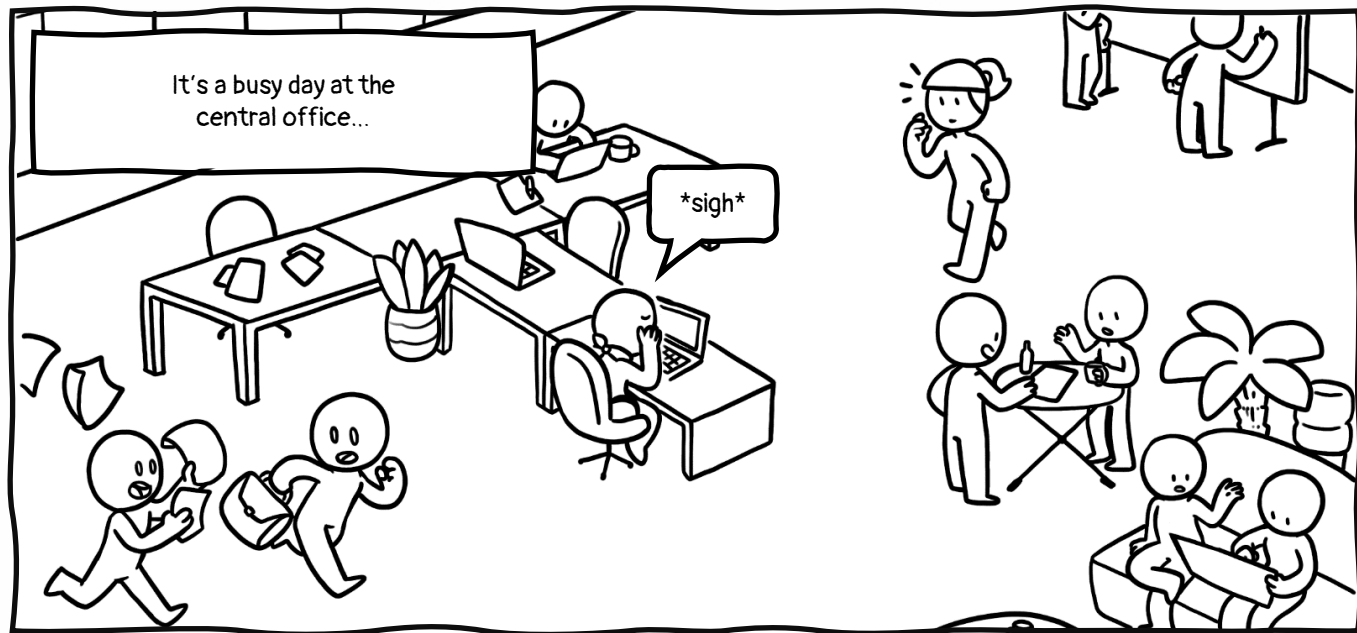
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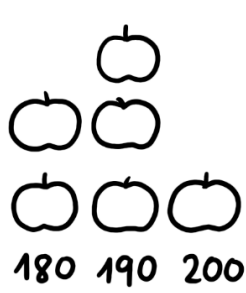
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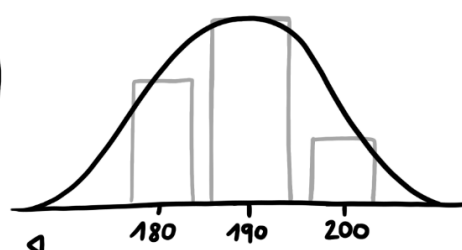


The best way to describe a variable is to report the values, and how often each value appears. That's called the **DISTRIBUTION** of the variable.



180 190 200

If we visualize the weight of our apples, the distribution would look like this, since all apples in this basket have around the same weight.

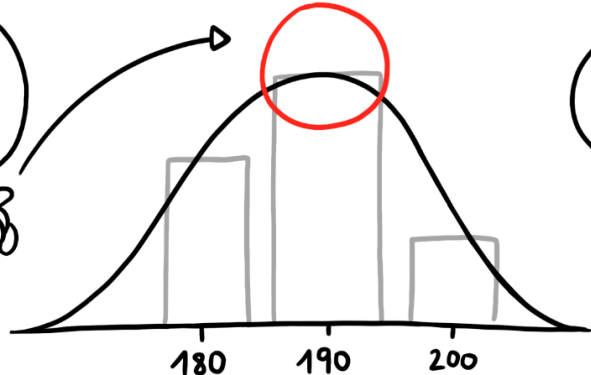


Weight of apples

It's kind of shaped like a bell...



Yes. If there were other apples in the basket, our best guess for their weight would be a value around that same spot.



Ah, okay. But... how would I calculate it?

This central point of our weight distribution represents well our data - that's why we call it the **CENTRAL TENDENCY**.

We can summarize the central tendency in multiple ways. The **MEAN** value is the most common way, and it's easy to calculate, too.



These are our six apples here:

200 180 190 190 190 180

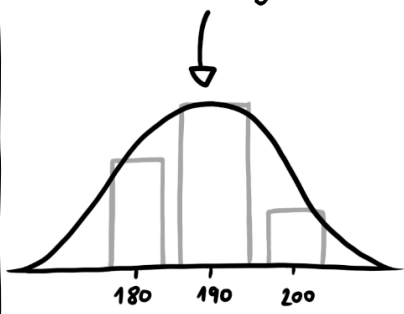
To calculate the mean, we add all the weight values together...

200 + 180 + 190 + 190 + 190 + 180

...and then divide by the number of apples we have...

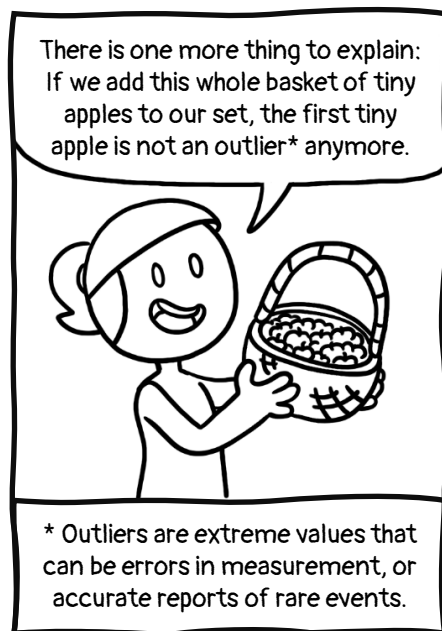
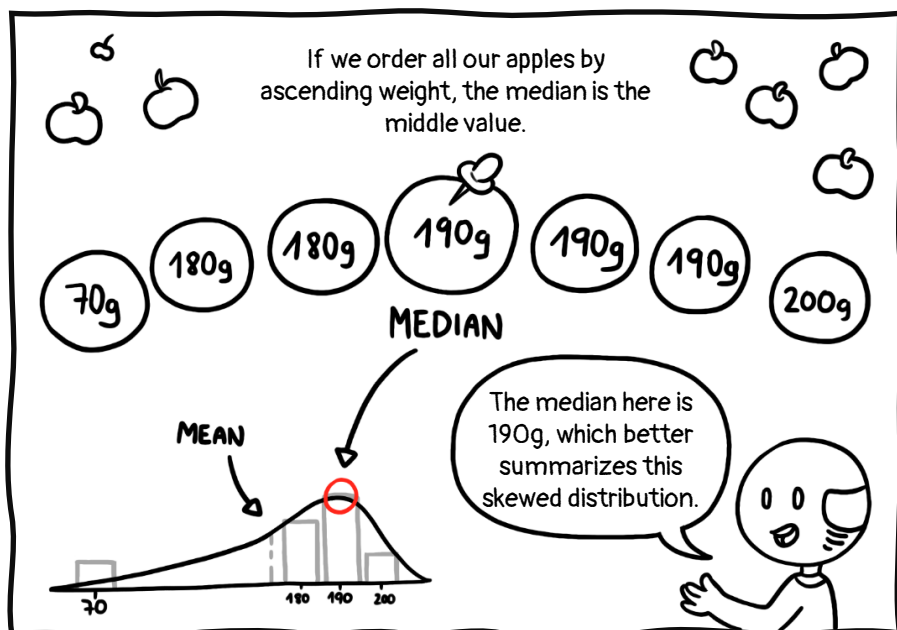
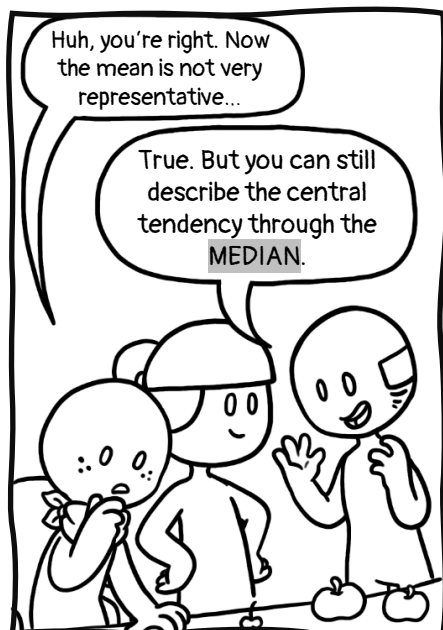
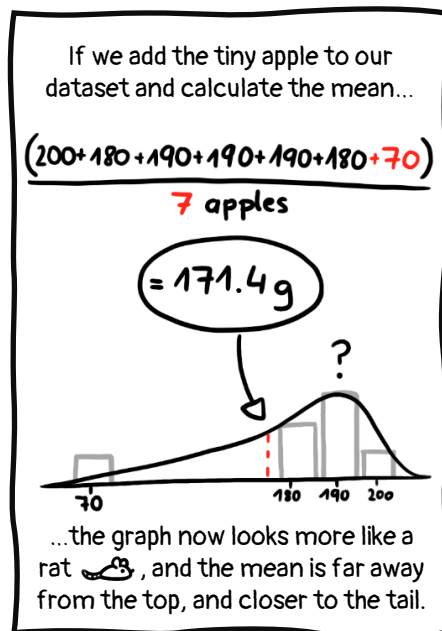
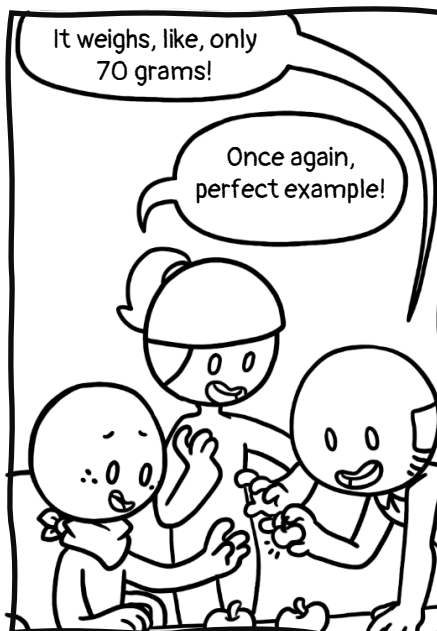
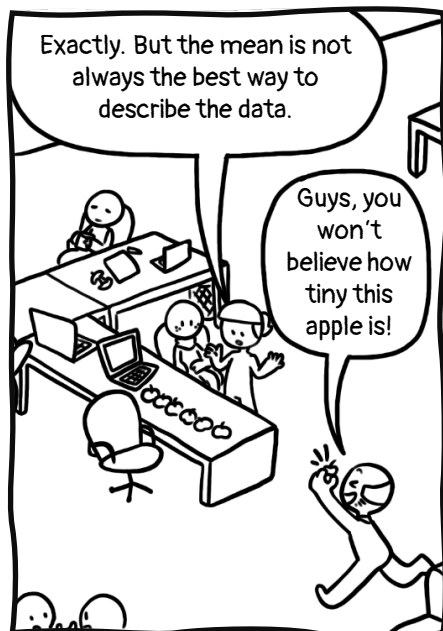
$$\frac{(200 + 180 + 190 + 190 + 190 + 180)}{6}$$

That gives us a mean value of 188.3g



Oh that's really almost the top of our bell-shaped graph!





In this case, our graph suddenly has two hills:



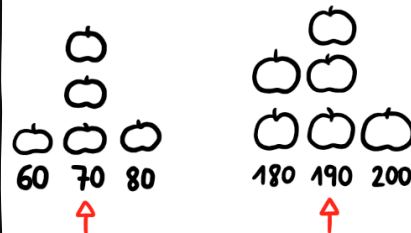
Like two rats in love!

Uhh, sure...

weight of tiny & big apples

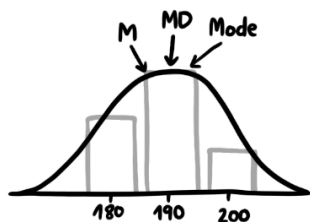
Anyway, we can use something called **MODE** to describe the central tendency if our distribution has multiple hills.

The mode defines the most frequently occurring value(s) in a dataset.



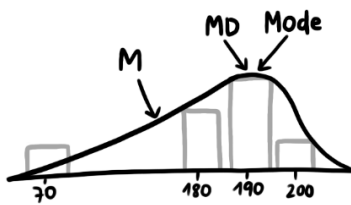
In this case, we have multiple modes, but there can also be just one, or even no mode at all.

You can apply mean, median, and mode to different samples of apples. But often, some will represent the data better than others.



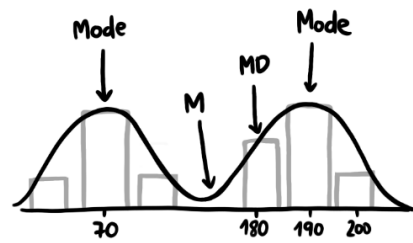
180, 180, 190, 190, 190, 200

$M = 188.3$ → good
 $MD = 190$ → parameters
 $Mode = 190$ →



70, 180, 181, 190, 191, 191, 200

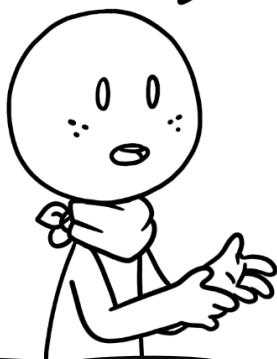
$M = 171.8$ → good
 $MD = 190$ → parameters
 $Mode = 191$ →



60, 70, 70, 70, 80, 180, 180, 190, 190, 190, 200

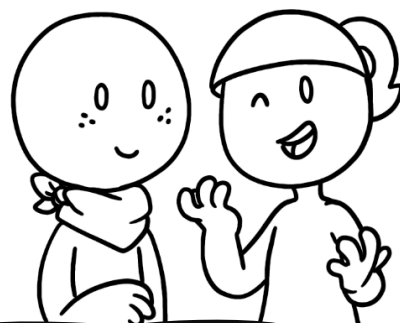
$M = 134.5$ → good
 $MD = 180$ → parameter
 $Mode = 70 \text{ \& } 190$ →

Okay, thank you... I've learned a lot. Now I just have to apply this to the data I have to present. It's from an app that tracks heart rate measurements.



User ID	Heart Rate (bpm)	Time of Use	User Rating
1	45	13:00	1
2	50	9:00	5
3	55	10:00	3
4	57	9:00	4
5	63	14:00	5
6	70	15:00	5
7	65	16:00	4
8	75	15:00	2

That should be doable - take a look at your data and follow the same steps we just did with the apples! You can use the next page for your calculations.





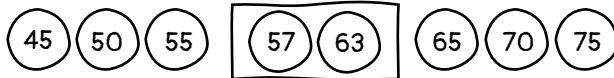
Before turning the page, try to calculate mean, median and mode for each variable, and check which parameter is most adequate for describing the central tendency. You can take notes on this page!

HEART RATE

Calculating the MEAN:

$$\frac{45+50+55+57+63+70+65+75}{8 \text{ users}} = \frac{480}{8} = 60 \text{ bpm}$$

Calculating the MEDIAN:



If there are two central values, the mean of the two values is the median:
 $(57+63)/2 = 60 \text{ bpm}$

Calculating the MODE:

45, 50, 55, 57, 63, 70, 65, 75

Each value only exists once -
 there is no mode!

If the distribution of the values is symmetrical, without any distortions, the mean is equal to the median.



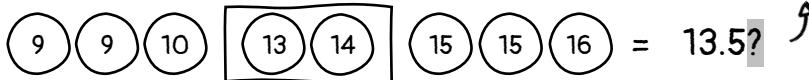
MOST FREQUENT TIME OF USE

Calculating the MEAN:

$$\frac{9+9+10+13+14+15+15+16}{8 \text{ users}} = \frac{101}{8} = 12.6?$$

Time of use is not a quantitative value - so calculating mean and median does not make any sense!

Calculating the MEDIAN:



Calculating the MODE:

9:00, 10:00, 13:00, 14:00, 15:00, 16:00
 2x 1x 1x 1x 2x 1x = 2 modes:
 9:00 & 15:00

Mode is not only suited for multimodal distributions, but also when working with ordinal and categorical data.



STAR RATING

Calculating the MEAN:

$$\frac{1+2+3+4+4+5+5+5}{8 \text{ users}} = \frac{29}{8} = 3.6 \text{ stars}$$

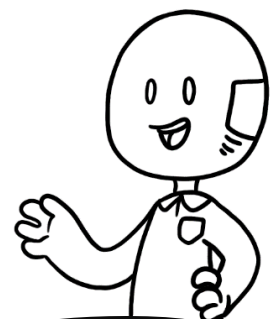
Calculating the MEDIAN:



Calculating the MODE:

1 2 3 4 5
 1x 1x 1x 2x 3x = 5 stars

For datasets with a skewed distribution, the median is a better way to describe central tendency.



Sources:

Downey, A. (2014). Think stats: exploratory data analysis. O'Reilly Media, Inc.

Field, A. (2022). An adventure in statistics: The reality enigma. Sage.

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